Omar Bassam

Cairo, Egypt | 01111474610 | omarbassam813@gmail.com | linkedin.com/in/omarelhediny | github.com/omarbemo

PROFESSIONAL SUMMARY

Computer Science student (Cairo University) — Machine Learning, Deep Learning, Computer Vision, GenAI/LLM fine-tuning, RAG, and agentic systems. Python, TensorFlow/Keras, Scikit-learn, XGBoost, FastAPI. Shipped real-world systems: sign language translation, fraud detection, anomaly detection, LLM ticket triage.

EDUCATION

Cairo University *Expected Graduation: 2027*

B.Sc. in Computer Science | GPA: 3.11

EXPERIENCE

Digital Egypt Pioneers Initiative (DEPI) – Ministry of Communications and IT *Nov 2025 – Jun 2026*

Microsoft Machine Learning Engineer | Team Lead

- Team Lead, 6-month Microsoft Azure ML/MLOps program — leading 5 engineers.
- Building real-time Two-Way Sign Language Translator (Computer Vision + Deep Learning).

NTI Summer Training Program *Sep 2026 – Oct 2026*

Cloud Architect Track

- 120-hour Cloud Architect training track.

IST Networks *Aug 2026 – Sep 2026*

Research & Development (R&D) Intern

- 4-week R&D internship — FastAPI, Git/GitHub, AI-assisted development, Deep Learning, cloud computing.
- Collaborated with R&D engineers on technical evaluation and research methodology.

Middle East Glass Manufacturing Company (MEG) *Jul 2026 (2 Weeks)*

Computer Vision Intern

- Built 3 production AI systems for glass factory floor deployment.
- Anomaly detection (Anomalib, ResNet18, 5 SOTA models, AUROC); FastAPI barcode+OCR pallet scanner; real-time YOLO monitoring pipeline.

COURSES

Data Science and AI Professional Track *Aug 2025 – May 2026*

AiTronix (Instructor: Ahmed Hafez)

- Full data science & ML pipeline — data wrangling, model building, statistical evaluation — NumPy, Pandas.

Machine Learning Course

CSKilled Academy

- End-to-end ML pipeline: preprocessing, modeling, evaluation, optimization.

PROJECTS

- **Two-Way Sign Language Translator – سوا —** *FastAPI, MediaPipe, TensorFlow/Keras*
- Real-time bidirectional ASL/ArASL translation; MediaPipe landmarks + TensorFlow/Keras classification; FastAPI live inference backend; 5-person team.
- **Credit Card Fraud Detection — End-to-End ML Pipeline** — *Scikit-learn, XGBoost, Keras, FastAPI, Docker*
- Benchmarked raw/undersampling/SMOTE/SMOTETomek across LR, Random Forest, XGBoost, Keras MLP; XGBoost champion model; packaged as tested FastAPI service — notebook to production.
- **LLM Fine-Tuning for Support Ticket Triage** — *Qwen3-4B, Unsloth, LoRA, TRL, FastAPI*
- Fine-tuned Qwen3-4B (LoRA/Unsloth) for queue/priority/type classification; Optuna hyperparameter search; TRL SFTTrainer; FastAPI inference service with unit-tested output parser.
- **Industrial Anomaly Detection & Pallet-Tracking System** — *Anomalib, ResNet18, PyTorch, FastAPI*
- 5 SOTA anomaly detection algorithms on ResNet18, ranked by AUROC; FastAPI barcode+OCR cross-validation pallet tracker.
- **Applied Data Science & Predictive Modeling Portfolio** — *Python, Scikit-Learn, Pandas, Seaborn*
- 3+ EDA & predictive modeling projects; IQR outlier detection & feature engineering (2,200+ records); 4+ ML algorithms compared via accuracy/precision/recall/F1.

TECHNICAL SKILLS

- **Core Programming & Languages:** Python, C++, Java, SQL, JavaScript, HTML/CSS
- **Data Science & AI/ML:** Machine Learning, Deep Learning, Computer Vision/Image Processing, Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Power BI
- **GenAI & MLOps:** LLM Fine-Tuning (LoRA/Unsloth/TRL), Prompt Engineering, FastAPI, Docker, Optuna
- **Computer Science Concepts:** Software Engineering Principles (SOLID), OOP, Data Structures & Algorithms, Design Patterns, ERD Modeling, Low-Code/No-Code Platforms
- **Design & Architecture Tools:** Figma, Draw.io
- **Soft Skills:** Collaboration, Analytical Problem Solving, Technical Communication, Research